

Deep Learning Practice Set

(All notations follow those used in the lecture slides)

*Learning rate is given as η

Section D: Recurrent Neural Networks (RNN)

1. Elman Network:

$$\mathbf{x}_t = [1, 0]^\top, \quad \mathbf{h}_{t-1} = [0.5, -0.5]^\top$$
$$W_{xh} = \begin{bmatrix} 0.2 & 0.3 \\ 0.4 & 0.1 \end{bmatrix}, \quad W_{hh} = \begin{bmatrix} 0.5 & -0.2 \\ 0.1 & 0.6 \end{bmatrix}, \quad W_y = \begin{bmatrix} 0.2 \\ 0.5 \end{bmatrix}$$

Compute the hidden state output h_t and the prediction y_t . Consider the bias terms as 0.

2. RNN Forward Pass

$$\mathbf{x}_t = [1, 0]^\top, \quad \mathbf{h}_{t-1} = [0.5, -0.5]^\top$$
$$W_{xh} = \begin{bmatrix} 0.2 & 0.3 \\ 0.4 & 0.1 \end{bmatrix}, \quad W_{hh} = \begin{bmatrix} 0.5 & -0.2 \\ 0.1 & 0.6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$

Compute the Hidden state output using tanh activation.

3. Elman Network (Two time steps)

Given:

$$\mathbf{x}_1 = [1, 0]^\top, \quad \mathbf{x}_2 = [0, 1]^\top, \quad \mathbf{h}_0 = [0, 0]^\top$$

$$W_{xh} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_{hh} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

Compute the hidden state outputs $\mathbf{h}_1$ and $\mathbf{h}_2$. Consider the bias terms as 0.

4. Bidirectional RNN: (f) denotes forward and (b) denotes backward

Given:

$$\mathbf{h}_t^{(f)} = [1, 2]^\top, \quad \mathbf{h}_t^{(b)} = [3, 4]^\top$$

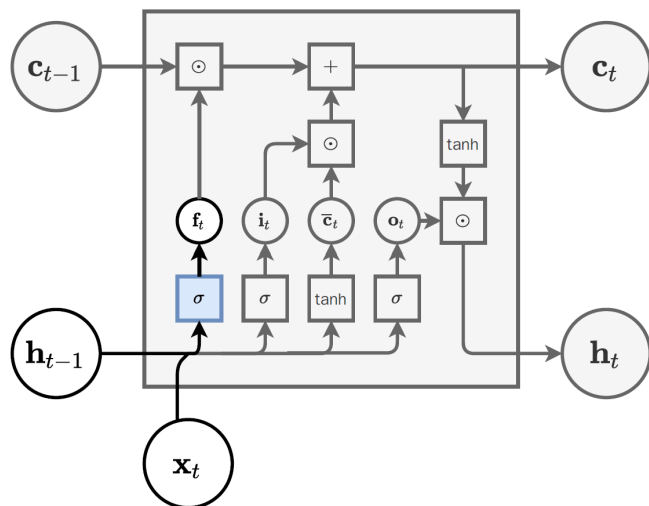
$$W = [0.5 \quad 0.2 \quad 0.1 \quad 0.3]^\top$$

Compute the concatenated hidden state output h_t . Also, compute the prediction y_t with tanh activation as given below.

$$\mathbf{y}_t = \tanh(W^\top \mathbf{h}_t)$$

5. Consider a simple LSTM cell with the following:

- **Input vector at time step t :** $\mathbf{x}_t = [1, 2]^\top$
- **Previous output state:** $\mathbf{h}_{t-1} = [0.5, -0.5]^\top$
- **Previous cell state:** $\mathbf{c}_{t-1} = [0, 0]^\top$



All weight matrices and biases are initialized as follows:

- W_f, W_i, W_o, W_c : all elements are equal to 1
- b_f, b_i, b_o, b_c : all elements are equal to 0

Compute:

- The new cell state $\mathbf{c}_t$
- The new output state $\mathbf{h}_t$